XIAMEN UNIVERSITY MALAYSIA
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COM105 Engineering Physics (I)

Lecture 11: Thermal Properties of Matter

热特性

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In this chapter, you'll learn…

- how to relate the pressure, volume, and temperature of a gas.
- how the pressure and temperature of a gas are related to the kinetic energy of its molecules.
- how the heat capacities of a gas reveal whether its molecules are rotating or vibrating.
- how the speeds of molecules are distributed in a gas.
- what determines whether a substance is a gas, a liquid, or a solid.

Food for thought

- How does the speed of the molecules 分子 in the air above the frying pan compare with that of the ones in the rest of the kitchen?
- How do the atoms (micro) of a gas determine its temperature (macro) and pressure?
- How do the microscopic properties of matter determine its macroscopic properties?



Equations of State

- Quantities such as pressure (p), volume (V), temperature (T), and the amount of a substance (n) are **state variables** because they describe the state of the substance. 物质
- The **state variables** depend only on the state of matter/system and not depend on how it is transforming.
- In a few cases the relationship among p, V, T, and m_{total} (or n) is simple enough that we can express it as an equation called the **equation of state**. num of moles
- When it's too complicated for that, we can use graphs or numerical tables.
- Even then, the relationship among the variables still exists.
- Gas is considered as a simple state of matter, we can get its equation-of-state via:
  - The ideal-gas equation at low pressure
  - The van der Wales Equation at high pressure

The Ideal-Gas Equation

- In the year 1800 – 1900s, scientists conducted many experiments on different gases.
- The ideal-gas equation is an equation of state for an ideal gas:

Ideal-gas equation:

$$pV = nRT \quad (18.3)$$

Gas pressure, Gas volume; Number of moles of gas, Absolute temperature of gas (K), Gas constant

- The universal gas constant R = 8.314 J/mol.K or 0.08206 L.atm/mol.K
- The molar mass M (molecular weight) is the mass per mole. The total mass of n moles is $m_{total} = nM$.

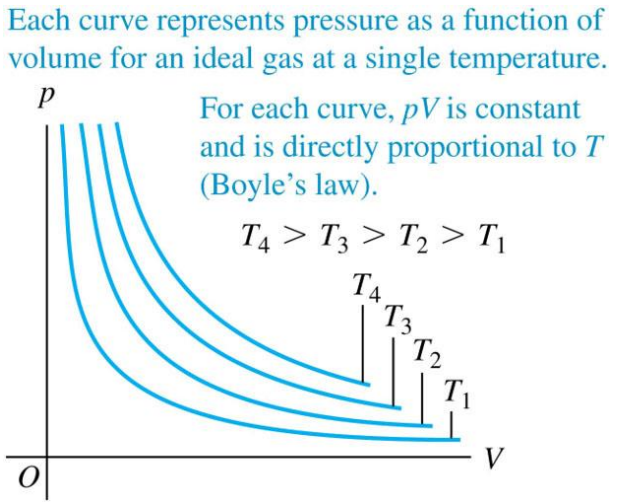
$$pV = \frac{m_{\text{total}}}{M}RT \quad (18.4)$$

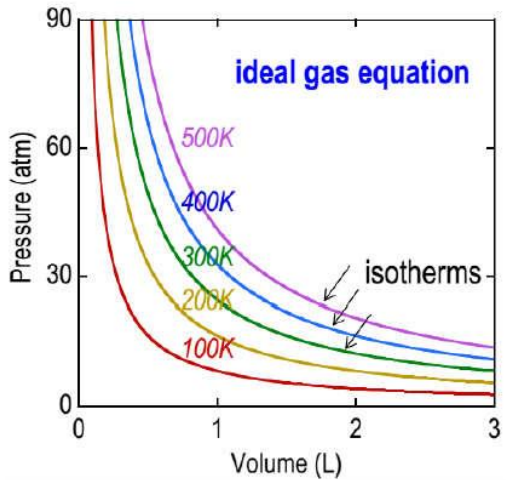
- From this we can get an expression for the density $\rho = m_{total}/V$ of the gas:

$$\rho = \frac{pM}{RT} \quad (18.5)$$

The Ideal-Gas Equation

- These show isotherms (等温线), or constant-temperature (pV) curves, for a constant amount of an ideal gas.





- The ideal gas equation is good for low pressure, ideal gas with constant mass:

$$\frac{p_1V_1}{T_1} = \frac{p_2V_2}{T_2} = \text{constant} \qquad \text{(ideal gas, constant mass)} \qquad (18.6)$$

The Ideal-Gas Equation

Example 1

You fill a rigid metal container with a volume of 20.0 L with nitrogen gas to a pressure of 197 atm at 28 °C. How many moles of gas are present in the container?

- P = 197 atm
- V = 20.0 L
- n = ?
- T = 28.0°C + 273 = 301 K

$PV = nRT$

$197\ \text{atm} \times 20.0\ \text{L} = n \times 0.082\ (28^\circ\text{C} + 273)\ \text{K}$

$\Rightarrow n = 159\ \text{mol}$

$$PV = nRT$$

$$(197\ \text{atm})(20.0\ \text{L}) = (n)(0.0821\ \text{L·atm/mol·K})(301\ \text{K})$$

$$n = 159\ \text{mol}$$

The Ideal-Gas Equation

Example 2

Calculate the pressure exerted by 50.0 g ethane, C_2H_6 in a 25.0 L container at 25.0 °C. Molar mass of ethene = 30.07 g/mol

- P = ?
- V = 25.0 L
- n = (50.0 g)(1 mol / 30.0 g) = 1.67 mol
- T = 25 °C + 273 = 298 K

$PV = nRT$

$P = \frac{50}{30.07}\ \text{mol} \times 0.082 \times (25+273)\ \text{K} \Big/ 25\ \text{L}$

$= 1.63\ \text{atm}$

$$PV = nRT$$

$$(P)(25.0\ \text{L}) = (1.67\ \text{mol})(0.0821\ \text{L·atm/mol·K})(298\ \text{K})$$

$$P = 1.63\ \text{atm}$$

The Ideal-Gas Equation

Example 3

The volume of a gas filled balloon is 30.0 L at 40 °C and 1148 mm Hg (millimeters of mercury) of pressure. What volume will the balloon have at 273 K and 1 atm (760 mm Hg)?

- P_1 = 1148 mm Hg
- V_1 = 30.0 L
- T_1 = 40.0°C = 313 K
- P_2 = 1 atm = 760 mm Hg
- V_2 = ?
- T_2 = 0°C = 273 K

30 L 40°C 1148 mm Hg.

273 K 1 atm = 760 mm Hg

$$P_1V_1T_2 = P_2V_2T_1$$

$$(1148\ \text{mm Hg})(30.0\ \text{L})(273\ \text{K}) = (760\ \text{mm Hg})(V_2)(313\ \text{K})$$

$$V_2 = 39.5\ \text{L}$$

$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$ $\frac{1148 \times 30}{273+40} = \frac{760 \times V_2}{273}$

$\Rightarrow V_2 =$

The Van Der Waals Equation

$PV = nRT$

- The model used for the ideal-gas equation ignores the volumes of molecules and the attractive forces between them. 忽略分子体积和吸引力
- The van der Waals equation is a more realistic model and is typically used in high pressure conditions:

Real gas molecules do interact with each other (they usually experience attraction at low pressures and repulsion at high pressures), hence different compressibility than ideal gases.

$p + \frac{an^2}{V^2}$ > 理论上压强

$$\left(p + \frac{an^2}{V^2}\right)(V - nb) = nRT \quad (18.7)$$

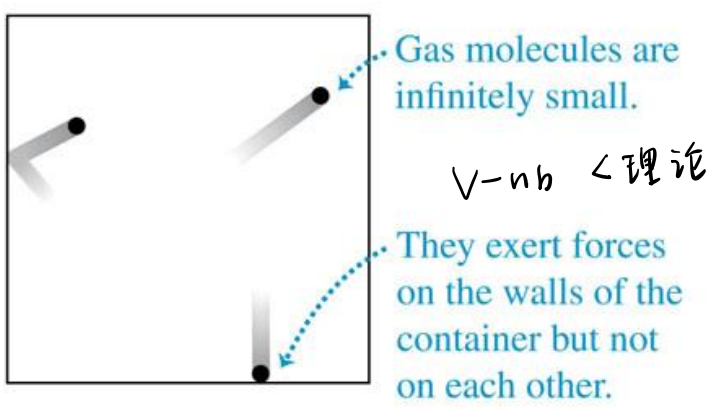
a depends on the attractive intermolecular forces, which reduce the pressure of the gas by *pulling* the molecules together as they *push* on the walls of the container.

b represents the volume of a mole of molecules

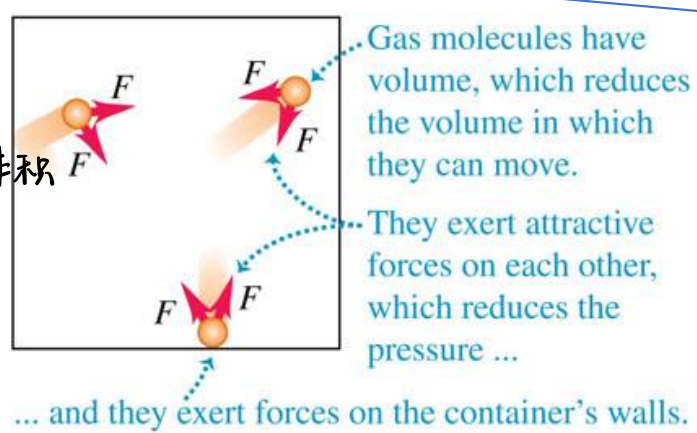
n number of moles

$V - nb$ < 理论上体积

To account for the volume that a real gas molecule takes up, the Van der Waals equation replaces V in the ideal gas law with (V - nb)



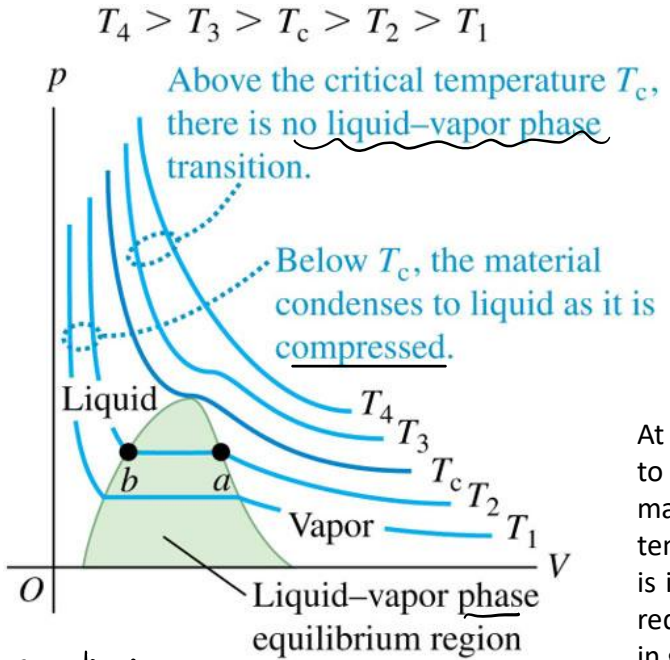
(a) An idealized model of a gas



(b) A more realistic model of a gas

The Van Der Waals Equation

- A pV-curves for a nonideal gas shows isotherms for temperatures above and below the critical temperature T_c.
- At still lower temperatures the material might undergo phase transitions from liquid to solid or from gas to solid; these are not shown here.



实际上的图像长这样

The flat parts of the isotherms in the shaded area represent conditions of liquid-vapor phase equilibrium. As the volume decreases, more and more material goes from vapor to liquid, but the pressure does not change. (To keep the temperature constant during condensation, we have to remove the heat of vaporization.)

At temperatures below Tc the isotherms develop flat regions in which we can compress the material (that is, reduce the volume V) without increasing the pressure p.

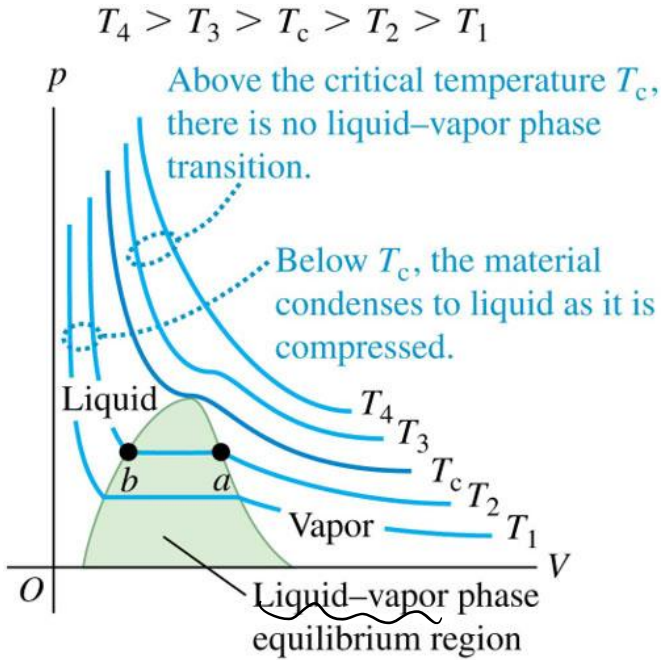
At T1 , similar behavior occurs, but the condensation begins at lower pressure and greater volume than at the constant temperature T2 .

At T2, it is vapor until point a is reached. Then it begins to liquefy; as the volume decreases further, more material liquefies, and both the pressure and the temperature remain constant. At point b, all the material is in the liquid state. After this, any further compression requires a very rapid rise of pressure, because liquids are in general much less compressible than gases.

phase change.

The Van Der Waals Equation

- A pV-curves for a nonideal gas shows isotherms for temperatures above and below the critical temperature T_c.
- At still lower temperatures the material might undergo phase transitions from liquid to solid or from gas to solid; these are not shown here.



At temperatures greater than Tc , no phase transition occurs as the material is compressed; at the highest temperatures, such as T4 , the curves resemble the ideal-gas curves

Molecular Properties of Matter 物质的分子性质

- Molecular Properties

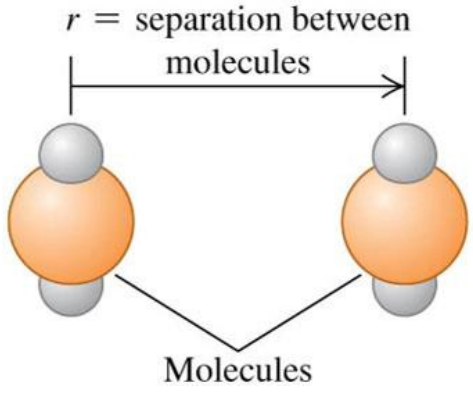
宏观 Macroscopic (P, V, T, m_{total}) (bulk behavior)

微观 Microscopic (Molecule, atom) Molecular Structure

化合物 相同 分子

- Any specific chemical compound is made up of identical molecules.
- In gases the molecules move nearly independently. move independently
- The force between molecules in a gas varies with the distance r between molecules.
- When molecules are far apart, the intermolecular forces are very small and usually attractive.

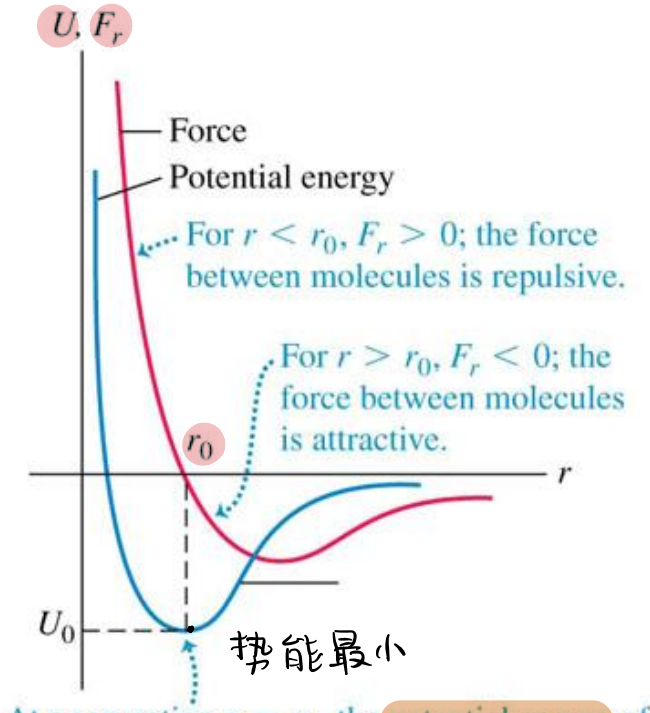
相吸



Molecules and Intermolecular Forces

分子间作用力

- The molecules in liquids and solids are held together by intermolecular forces, arise from interactions among the electrically charged particles that make up the molecules.
- Molecules in solids are essentially fixed in place, while those in liquids and gases have much more freedom to move.
- The force between molecules in a gas varies with the distance r between molecules.
- Positive F_r corresponds to a repulsive force 斥力 and a negative F_r to an attractive force. 引力
- A molecule at rest at a distance r_0 from a second molecule would need an additional energy $|U_0|$, the "depth" of the potential well, to "escape" to an indefinitely large value of r.

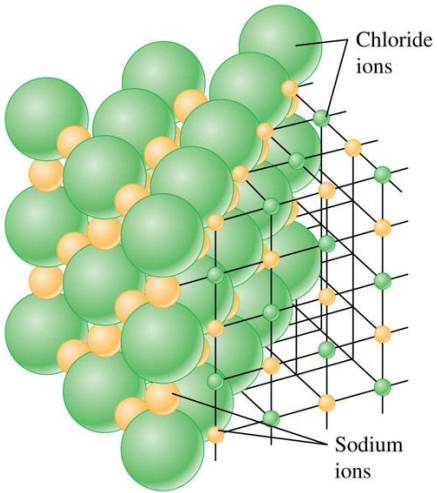


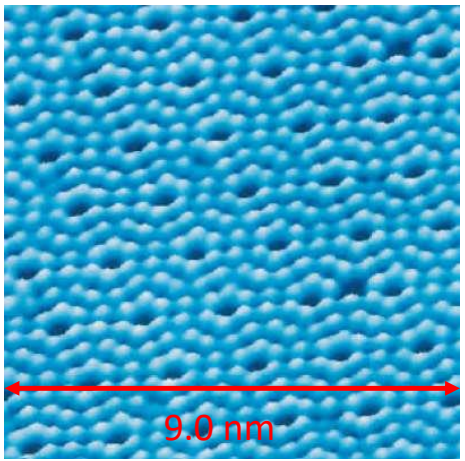
How the force between molecules and their interaction potential energy depend on their separation r.

Molecules and Intermolecular Forces

- In a crystalline solid, these points are arranged in a crystal lattice.
- In a liquid, the intermolecular distances are usually only slightly greater than in the solid phase of the same substance, but the molecules have much greater freedom of movement.
- Liquids show regularity of structure only in the immediate neighborhood of a few molecules.

This is the schematic representation of the cubic crystal structure of sodium chloride NaCl (ordinary salt).

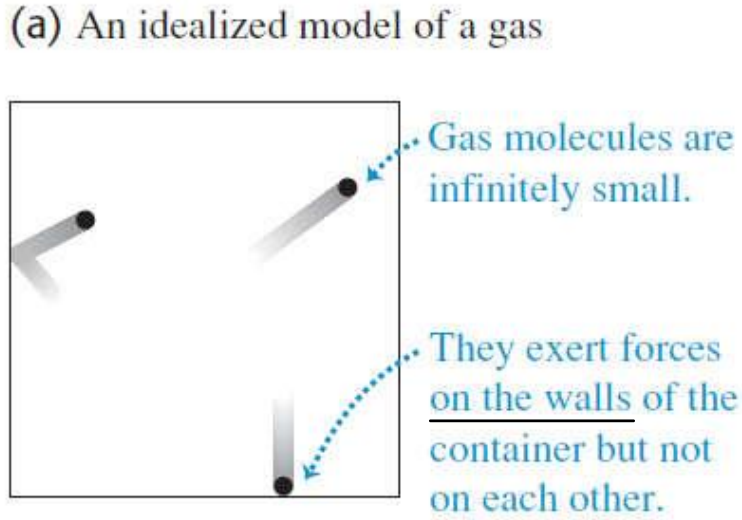




This is a scanning tunneling microscope image of the surface of a silicon crystal. Each blue "bead" is one silicon atom; these atoms are arranged in a (nearly) perfect array of hexagons.

Molecules and Intermolecular Forces

- In gases the molecules move nearly independently.
- The molecules of a gas are usually widely separated and so have only very small attractive forces. A gas molecule moves in a straight line until it collides with another molecule or with a wall of the container.
- An ideal gas, in molecular terms, is a gas whose molecules exert no attractive forces on each other and therefore have no potential energy.



Moles and Avogadro's Number 阿佛加德罗常数

- One mole of a substance contains as many elementary entities (atoms or molecules) as there are atoms in 0.012 kg of carbon-12.
- One mole of a substance contains Avogadro's number N_A of molecules.

  $N_A = 6.022 \times 10^{23}$ molecules/mol
- The molar mass M is the mass of one mole.



(18.8)

- When the molecule consists of a single atom, the term atomic mass is often used instead of molar mass.

$N = n N_A$

↑ number of moles.

Moles and Avogadro's Number

$M_H = 1.008\ g/mol = N_A \cdot m$

$m_H = \frac{1.008\ g/mol}{6.022 \times 10^{23}\ atoms/mol} = 1.674 \times 10^{-24}\ g/atom$

$m_{O_2} = \frac{32\ g/mol}{6.022 \times 10^{23}\ molecules/mol} = 53.1 \times 10^{-2}\ g/molecule$

EXAMPLE 18.5 ATOMIC AND MOLECULAR MASS

Find the mass of a single hydrogen atom and of a single oxygen molecule.

SOLUTION

IDENTIFY and SET UP: This problem involves the relationship between the mass of a molecule or atom (our target variable) and the corresponding molar mass M. We use Eq. (18.8) in the form $m = M/N_A$ and the values of the atomic masses from the periodic table of the elements (see Appendix D).

EXECUTE: For atomic hydrogen the atomic mass (molar mass) is $M_H = 1.008$ g/mol, so the mass m_H of a single hydrogen atom is

$$m_H = \frac{1.008\ \text{g/mol}}{6.022 \times 10^{23}\ \text{atoms/mol}} = 1.674 \times 10^{-24}\ \text{g/atom}$$

For oxygen the atomic mass is 16.0 g/mol, so for the diatomic (two-atom) oxygen molecule the molar mass is 32.0 g/mol. Then the mass of a single oxygen molecule is

$$m_{O_2} = \frac{32.0\ \text{g/mol}}{6.022 \times 10^{23}\ \text{molecules/mol}} = 53.1 \times 10^{-24}\ \text{g/molecule}$$

EVALUATE: We note that the values in Appendix D are for the *average* atomic masses of a natural sample of each element. Such a sample may contain several *isotopes* of the element, each with a different atomic mass. Natural samples of hydrogen and oxygen are almost entirely made up of just one isotope.

Kinetic-Molecular Model of an Ideal Gas

- A simple molecular model of an ideal gas that represents the gas as a large number of particles bouncing around in a closed container.
- The assumptions of the kinetic-molecular model are: 动力学分子模组

1. A container contains a very large number of N of identical molecules each with mass m.
2. The molecules behave like point particles that are small compared to the size of the container and the average distance between molecules.
3. The molecules are in constant motion and undergo perfectly elastic collisions. 完全弹性碰撞
4. The container walls are perfectly rigid and do not move.

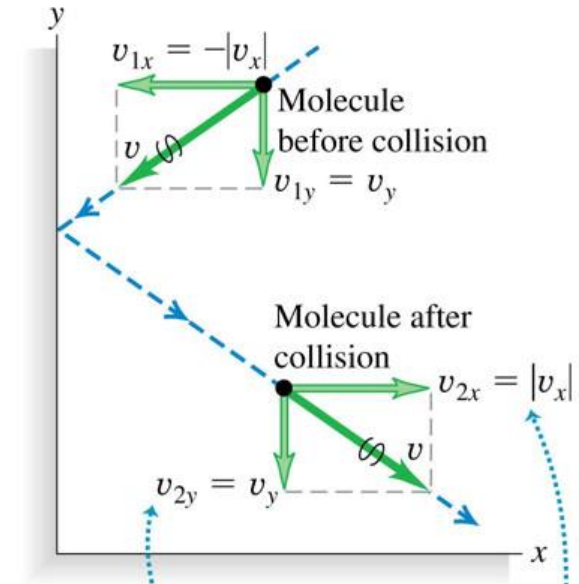
碰撞与气体压强

Collisions and Gas Pressure

$P = \frac{F}{A}$

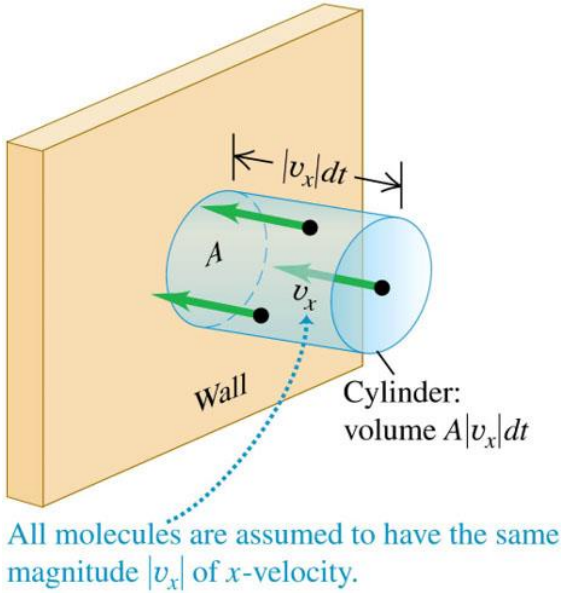
- During collisions the molecules exert forces on the walls of the container; this is the origin of the pressure that the gas exerts.
- In a typical collision (shown) the velocity component parallel to the wall is unchanged, and the component perpendicular to the wall reverses direction but does not change in magnitude.

垂直



Collisions and Gas Pressure

- If a molecule is going to collide with a given wall area *A* during a small time interval *dt*, it must be within a distance $|v_x|dt$ from the wall (shown) and it must be headed toward the wall.
- So the number of molecules that collide with A during *dt* is equal to the number of molecules within the cylinder that have their x-velocity aimed toward the wall.



- Total momentum changed in *dt*,

$$dP_x = \tfrac{1}{2}\left(\frac{N}{V}\right)(A|v_x|\,dt)(2m|v_x|) = \frac{NAmv_x^2\,dt}{V} \qquad (18.9)$$

$$\frac{dP_x}{dt} = \frac{NAmv_x^2}{V} \qquad (18.10)$$

- The rate of change of momentum equals the force exerted by the wall area A on the gas molecules

$$p = \frac{F}{A} = \frac{Nmv_x^2}{V} \qquad (18.11)$$

压强

$P = mv$ $\quad F = ma = m\frac{v}{t} = \frac{P}{t}$

Pressure and Molecular Kinetic Energies

- The average speed of a molecule is related to the velocity components *Vx, Vy,* and *Vz*, by

$$(v^2)_{av} = (v_x^2)_{av} + (v_y^2)_{av} + (v_z^2)_{av}$$

- There is no real difference in our model between the x-, y-, and z-directions (Molecular speeds are very fast in a typical gas, so the effects of gravity are negligibly small.)

$$(v_x^2)_{av} = (v_y^2)_{av} = (v_z^2)_{av} \quad \Rightarrow \quad (v_x^2)_{av} = \tfrac{1}{3}(v^2)_{av}$$

- Hence, Eq 18.11 becomes

$$p = \frac{F}{A} = \frac{Nmv_x^2}{V} \quad \Rightarrow \quad pV = \tfrac{1}{3}Nm(v^2)_{av} = \tfrac{2}{3}N\left[\tfrac{1}{2}m(v^2)_{av}\right] \qquad (18.12)$$

$\frac{1}{2}m(v^2)_{av}$ The average kinetic translational kinetic energy for one single molecule.

$N\left[\frac{1}{2}m(v^2)_{av}\right]$ The energy for translational motion of all the molecules

The product *pV* equals two-thirds of the total translational kinetic energy

$$pV = \tfrac{2}{3}K_{tr} \qquad (18.13)$$

$pV = nmv_x^2$ = 用 v_{av}

N = very large number of identical molecules

Pressure and Molecular Kinetic Energies

- Since $pV = nRT$ and $pV = \frac{2}{3}K_{\text{tr}}$
- The total random kinetic energy K_{tr} of translational motion of all the molecules in a gas is directly proportional to the absolute temperature T:

Average translational kinetic energy of an ideal gas

$$K_{\text{tr}} = \frac{3}{2}nRT$$

- n: Number of moles of gas
- T: Absolute temperature of gas
- R: Gas constant

(18.14)

- Then $pV = \frac{1}{3}Nm(v^2)_{\text{av}} = \frac{2}{3}N\left[\frac{1}{2}m(v^2)_{\text{av}}\right]$

$$\frac{K_{\text{tr}}}{N} = \frac{1}{2}m(v^2)_{\text{av}} = \frac{3nRT}{2N}$$

$N = nN_A$ $\quad \frac{n}{N} = \frac{1}{N_A}$

$$\frac{K_{\text{tr}}}{N} = \frac{1}{2}m(v^2)_{\text{av}} = \frac{3}{2}\left(\frac{R}{N_A}\right)T$$

(18.15)

N = very large number of identical molecules

- The ratio R/N_A is called the Boltzmann constant, k:

$$k = \frac{R}{N_A} = \frac{8.314\ \text{J/mol} \cdot \text{K}}{6.022 \times 10^{23}\ \text{molecules/mol}} = 1.381 \times 10^{-23}\ \text{J/molecule} \cdot \text{K}$$

Pressure and Molecular Kinetic Energies

- This means that the average translational kinetic energy per molecule K_{tr}/N depends only on the temperature, not on the pressure, volume, or kind of molecule.

Average translational kinetic energy of a gas molecule

$$\tfrac{1}{2}m(v^2)_{\text{av}} = \tfrac{3}{2}kT \qquad (18.16)$$

- m: Mass of a molecule
- $(v^2)_{\text{av}}$: Average value of the square of molecular speeds
- T: Absolute temperature of gas
- k: Boltzmann constant
- N = very large number of identical molecules

$N = nN_A$

n = number of moles

M = molar mass, mass of one mole

$M = N_A m$ (18.8)

m = mass of a molecule of substance

$k = \dfrac{R}{N_A}$

$$N_A \tfrac{1}{2}m(v^2)_{\text{av}} = \tfrac{1}{2}M(v^2)_{\text{av}} = \tfrac{3}{2}RT$$

(average translational kinetic energy per mole of gas) (18.17)

- This shows that the translational kinetic energy of a mole of an ideal gas depends only on T.

$pV = nRT$ ⇓ $N = N_A n$ $R = N_A k$

$$pV = NkT \qquad (18.18)$$

Pressure and Molecular Kinetic Energies

- This shows that we can think of the Boltzmann constant *k* as a gas constant on a "per-molecule" basis instead of the usual "per-mole" basis for *R*.

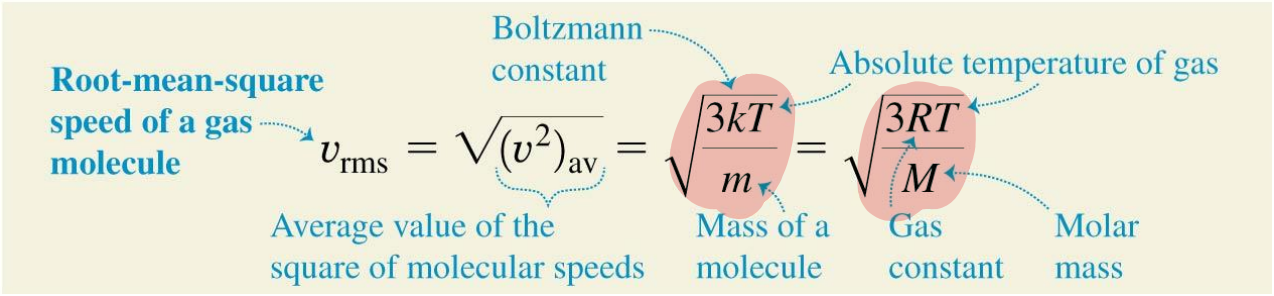
Summary for the ideal-gas equation

$$pV = nRT \qquad \text{(ideal-gas equation)}$$

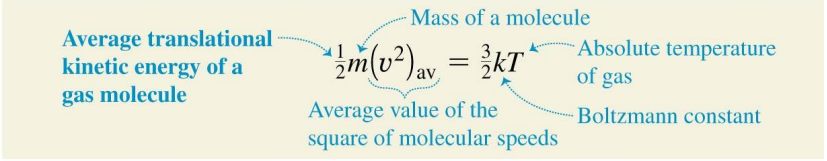
$$pV = NkT \qquad (k\text{: Boltzmann Constant})$$

Pressure and Molecular Kinetic Energies

- To compute the rms speed, we square each molecular speed, add, divide by the number of molecules, and take the square root; v_{rms} is the root of the mean of the squares.
- Expressions for the square root of $(v^2)_{av}$, called the root-mean-square speed of the molecules in a gas is:



(18.19)



Pressure and Molecular Kinetic Energies

EXAMPLE 18.6 MOLECULAR KINETIC ENERGY AND v_{rms}

$\frac{1}{2}m(v^2)_{av} = \frac{3}{2}kT = \frac{3}{2}(1.38\times10^{-23}\,J/K)$

(a) What is the average translational kinetic energy of an ideal-gas molecule at 27°C? (b) What is the total random translational kinetic energy of the molecules in 1 mole of this gas? (c) What is the rms speed of oxygen molecules at this temperature?

$\frac{1}{2}m(v^2)_{av}$

K_{tr}

EXECUTE: (a) From Eq. (18.16),

$$\frac{1}{2}m(v^2)_{av} = \frac{3}{2}kT = \frac{3}{2}(1.38 \times 10^{-23}\ \text{J/K})(300\ \text{K})$$
$$= 6.21 \times 10^{-21}\ \text{J}$$

(b) From Eq. (18.14), the kinetic energy of one mole is

$$K_{tr} = \frac{3}{2}nRT = \frac{3}{2}(1\ \text{mol})(8.314\ \text{J/mol}\cdot\text{K})(300\ \text{K}) = 3740\ \text{J}$$

(c) We found the mass per molecule m and molar mass M of molecular oxygen in Example 18.5. Using Eq. (18.19), we can calculate v_{rms} in two ways:

$$v_{rms} = \sqrt{\frac{3kT}{m}} = \sqrt{\frac{3(1.38 \times 10^{-23}\ \text{J/K})(300\ \text{K})}{5.31 \times 10^{-26}\ \text{kg}}}$$
$$= 484\ \text{m/s} = 1740\ \text{km/h} = 1080\ \text{mi/h}$$

$$v_{rms} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3(8.314\ \text{J/mol}\cdot\text{K})(300\ \text{K})}{32.0 \times 10^{-3}\ \text{kg/mol}}} = 484\ \text{m/s}$$

Pressure and Molecular Kinetic Energies

EXAMPLE 18.7 CALCULATING RMS AND AVERAGE SPEEDS

Five gas molecules have speeds of 500, 600, 700, 800, and 900 m/s. What is the rms speed? What is the *average* speed?

EXECUTE: The average value of v^2 and the resulting v_{rms} for the five molecules are

$$(v^2)_{\text{av}} = \frac{500^2 + 600^2 + 700^2 + 800^2 + 900^2}{5}\ \text{m}^2/\text{s}^2$$

$$= 5.10 \times 10^5\ \text{m}^2/\text{s}^2$$

$$v_{\text{rms}} = \sqrt{(v^2)_{\text{av}}} = 714\ \text{m/s}$$

rms speed

The average speed v_{av} is

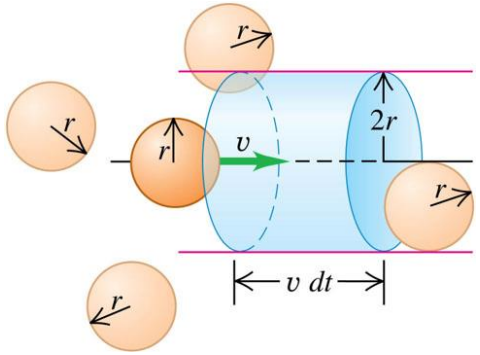
average speed

$$v_{\text{av}} = \frac{500 + 600 + 700 + 800 + 900}{5}\ \text{m/s} = 700\ \text{m/s}$$

之前不在意这个，ideal 不考虑

Collisions Between Molecules

- We have ignored the possibility that two gas molecules might collide.
- A more realistic model: N spherical molecules as rigid spheres of radius r in a volume V, with only one molecule is moving parallel with the axis of the cylinder (radius $2r$).

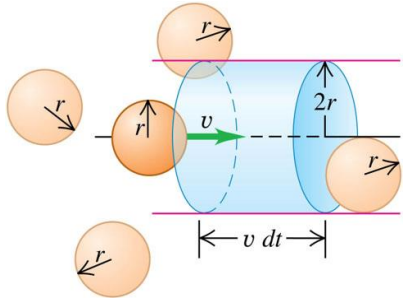


- In a short time dt a molecule with speed v travels a distance $v\,dt$; during this time it collides with any molecule that is in the cylindrical volume of radius $2r$ and length $v\,dt$. The volume of cylinder is: $4\pi r^2 v\,dt$
- There are N/V molecules per unit volume, so the number dN with centers in this cylinder is: $dN = 4\pi r^2 v\,dt\,N/V$
- Thus the number of collisions per unit time is: $\dfrac{dN}{dt} = \dfrac{4\pi r^2 v N}{V}$

Collisions Between Molecules

- This result assumes that only one molecule is moving. It turns out that collisions are more frequent when all the molecules move at once, and the above equation has to be multiplied by a factor of $\sqrt{2}$.

$$\frac{dN}{dt} = \frac{4\pi\sqrt{2}r^2vN}{V}$$

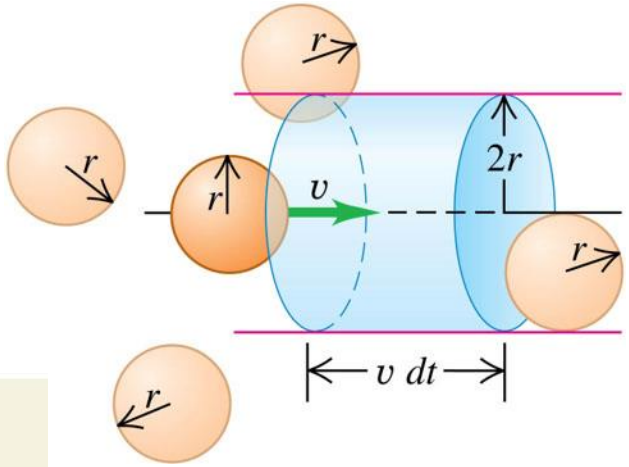


- The average time t_{mean} between collisions, called the mean free time, is the reciprocal of this expression:

$$t_{\text{mean}} = \frac{V}{4\pi\sqrt{2}r^2vN} \qquad (18.20)$$

Collisions Between Molecules

- The average distance traveled between collisions is called the mean free path.
- In our model, this is just the molecule's speed v multiplied by mean free time:



Mean free path of a gas molecule

$$\lambda = vt_{\text{mean}} = \frac{V}{4\pi\sqrt{2}r^2N} \qquad (18.21)$$

Speed of molecule; Volume of gas; Mean free time between collisions; Radius of a molecule; Number of molecules in gas

$$pV = NkT$$

$$\lambda = \frac{kT}{4\pi\sqrt{2}r^2p} \qquad (18.22)$$

- The more molecules there are and the larger the molecule, the shorter the mean distance between collisions.

Collisions Between Molecules

EXAMPLE 18.8 CALCULATING MEAN FREE PATH

(a) Estimate the mean free path of a molecule of air at 27°C and 1 atm. Model the molecules as spheres with radius $r = 2.0 \times 10^{-10}$ m. (b) Estimate the mean free time of an oxygen molecule with $v = v_{rms}$ at 27°C and 1 atm.

EXECUTE: (a) From Eq. (18.22),

$$\lambda = \frac{kT}{4\pi\sqrt{2}r^2p} = \frac{(1.38 \times 10^{-23}\ \text{J/K})(300\ \text{K})}{4\pi\sqrt{2}(2.0 \times 10^{-10}\ \text{m})^2(1.01 \times 10^{5}\ \text{Pa})}$$

$$= 5.8 \times 10^{-8}\ \text{m}$$

(b) From Example 18.6, for oxygen at 27°C the root-mean-square speed is $v_{rms} = 484$ m/s, so the mean free time for a molecule with this speed is

$$t_{mean} = \frac{\lambda}{v} = \frac{5.8 \times 10^{-8}\ \text{m}}{484\ \text{m/s}} = 1.2 \times 10^{-10}\ \text{s}$$

This molecule undergoes about 10^{10} collisions per second!

EXAMPLE 18.6

(c) We found the mass per molecule m and molar mass M of molecular oxygen in Example 18.5. Using Eq. (18.19), we can calculate v_{rms} in two ways:

$$v_{rms} = \sqrt{\frac{3kT}{m}} = \sqrt{\frac{3(1.38 \times 10^{-23}\ \text{J/K})(300\ \text{K})}{5.31 \times 10^{-26}\ \text{kg}}}$$

$$= 484\ \text{m/s} = 1740\ \text{km/h} = 1080\ \text{mi/h}$$

$$v_{rms} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3(8.314\ \text{J/mol}\cdot\text{K})(300\ \text{K})}{32.0 \times 10^{-3}\ \text{kg/mol}}} = 484\ \text{m/s}$$

$$k = \frac{R}{N_A} = \frac{8.314\ \text{J/mol}\cdot\text{K}}{6.022 \times 10^{23}\ \text{molecules/mol}} = 1.381 \times 10^{-23}\ \text{J/molecule}\cdot\text{K}$$

Heat Capacities of Gases

- The degrees of freedom are the number of velocity components needed to describe a molecule completely.
- A monatomic gas has three degrees of freedom and a diatomic gas has five; the equipartition of energy principle states that each degree of freedom has of ½ kT kinetic energy associated with it.
- For an ideal monatomic gas: 单原子 Ex: He.

Molar heat capacity at constant volume, ideal gas of point particles

$$C_V = \frac{3}{2}R$$ ← Gas constant (18.25)

- For an ideal diatomic gas: Ex: O_2, CO, H_2

二原子的

Molar heat capacity at constant volume, ideal diatomic gas

$$C_V = \frac{5}{2}R$$ ← Gas constant (18.26)

Heat Capacities of Gases

- Table shows that the measured values for C_V for monatomic gases and diatomic gases agree quite well with the calculated values.

測得 Measured values:

Type of Gas	Gas	C_v (j/mol·K)
Monatomic	He	12.47
	Ar	12.47
Diatomic	H_2	20.42
	N_2	20.76
	O_2	20.85
	CO	20.85
Polyatomic	CO_2	28.46
	SO_2	31.39
	H_2S	25.95

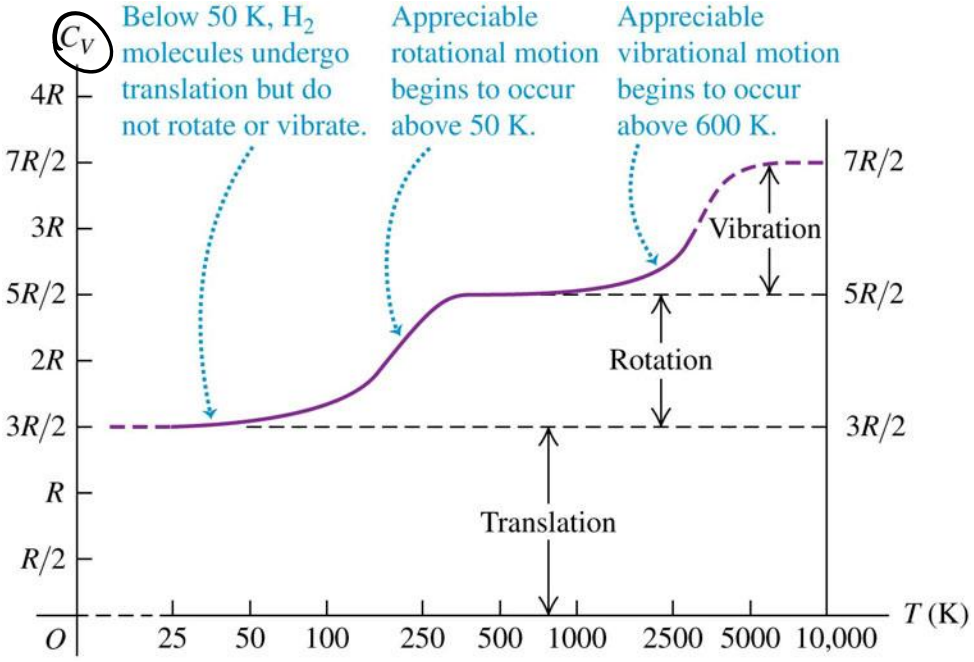
Calculated values:

$$C_V = \frac{3}{2}(8.314\ \text{J/mol}\cdot\text{K}) = 12.47\ \text{J/mol}\cdot\text{K}$$

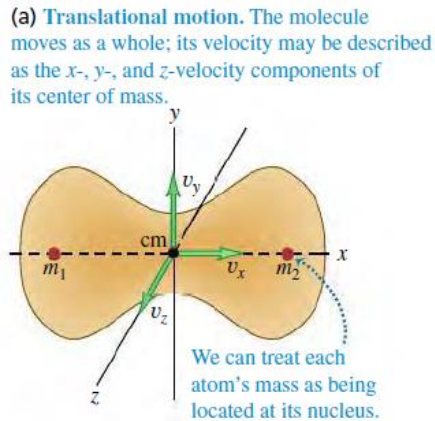
$$C_V = \frac{5}{2}(8.314\ \text{J/mol}\cdot\text{K}) = 20.79\ \text{J/mol}\cdot\text{K}$$

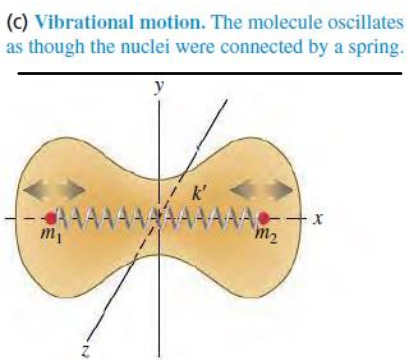
Heat Capacities of Gases

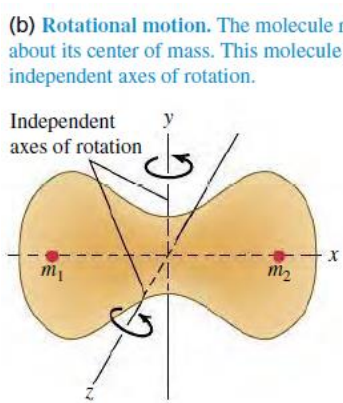
- Experimental Values of C_V for Hydrogen Gas (H_2)



Vibrational motion can also contribute to the heat capacities of gases. Molecular bonds can stretch and bend, and the resulting vibrations lead to additional degrees of freedom and additional energies. For most diatomic gases, however, vibration does *not* contribute appreciably to heat capacity.







Heat Capacities of Solids

- Consider a crystal consisting of N identical atoms (a monatomic solid).
- Each atom is bound to an equilibrium position by interatomic forces.
- We can think of a crystal as an array of atoms connected by little springs.
- Each atom has an average kinetic energy 3/2 kT and an average potential energy 3/2 kT, or an average total energy 3 kT per atom.
- The molar heat capacity of a crystal is:

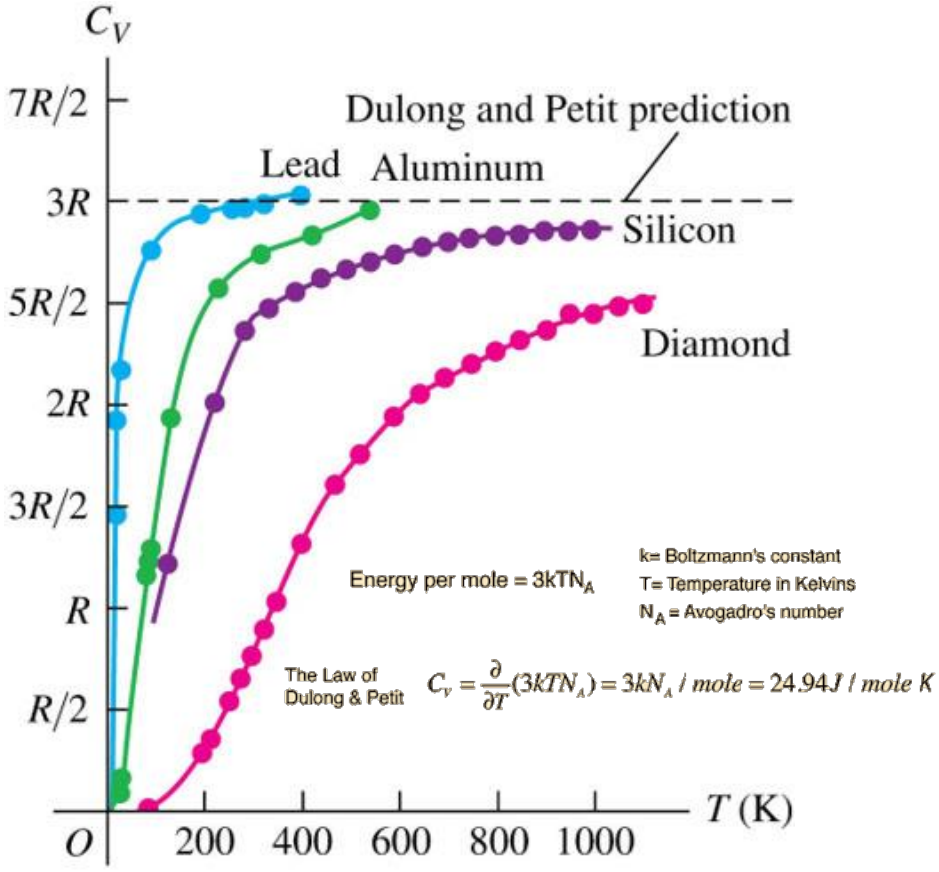
Molar heat capacity of an ideal monatomic solid (rule of Dulong and Petit)

$$C_V = 3R \quad \text{(Gas constant)} \qquad (18.28)$$

$$C_V = (3)(8.314\ \text{J/mol} \cdot \text{K}) = 24.9\ \text{J/mol} \cdot \text{K}$$

Heat Capacities of Solids

- Experimental values of C_V for lead, aluminum, silicon, and diamond are given in the figure.
- At high temperatures, C_V for each solid approaches about $3R$, in agreement with the rule of Dulong and Petit.
- At low temperatures, C_V is much less than $3R$.



分子速度 Molecular Speeds

- The molecules in a gas don't all have the same speed.
- The molecule speed can be measured by many advanced instruments.
- Theoretically, a distribution function *f(v)* is defined to describe the experimental results.
- For a total of *N* molecules, the number *dN* having speeds in the range between *v* and *v*+*dv* is given by

$$dN = Nf(v)\,dv \qquad (18.29)$$

- *f(v)dv* is the probability that a randomly chosen molecule that will have a speed in the interval *v* to *v*+*dv.*
- The distribution function *f(v)* is the probability per unit speed interval; it is not equal to the probability that a molecule has speed exactly equal to *v*.
- Since a probability is a pure number, *f(v)* has units of reciprocal speed (*s/m*).

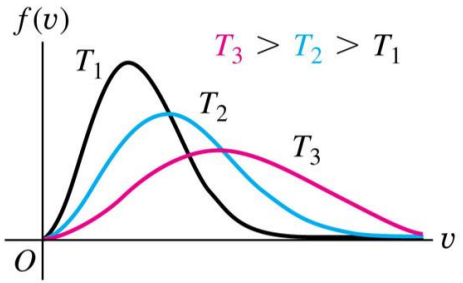
Molecular Speeds

- The average speed and the rms speed

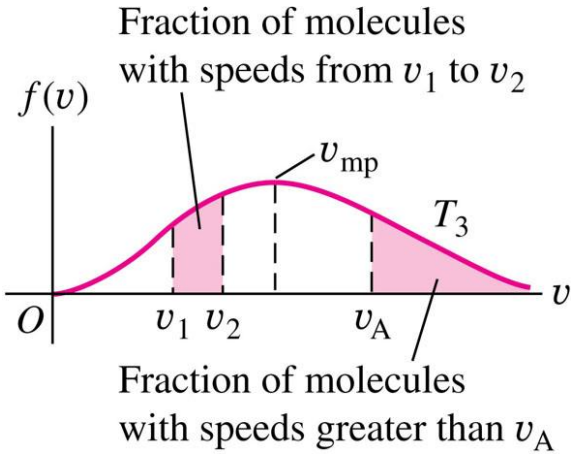
$$v_{\text{av}} = \int_0^\infty v f(v)\, dv \qquad (18.30)$$

$$(v^2)_{\text{av}} = \int_0^\infty v^2 f(v)\, dv \qquad (18.31)$$

- The Maxwell-Boltzmann distribution *f(v)* gives the distribution of molecular speeds.
- At each temperature the height of the curve for any value of *v* is proportional to the number of molecules with speeds near *v*. The peak of the curve represents the most probable speed v_{mp} for the corresponding temperature.

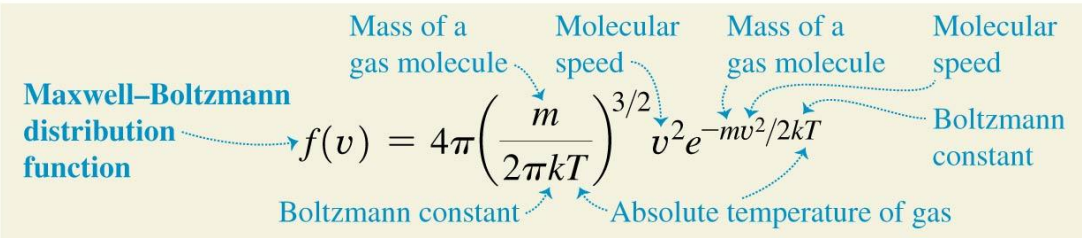


As temperature increases:
- the curve flattens.
- the maximum shifts to higher speeds.



Molecular Speeds

- The Maxwell–Boltzmann distribution *f(v)* describing the actual distribution of molecular speeds can be derived from statistical mechanics considerations, but that derivation is beyond our scope.
- Here is the result:

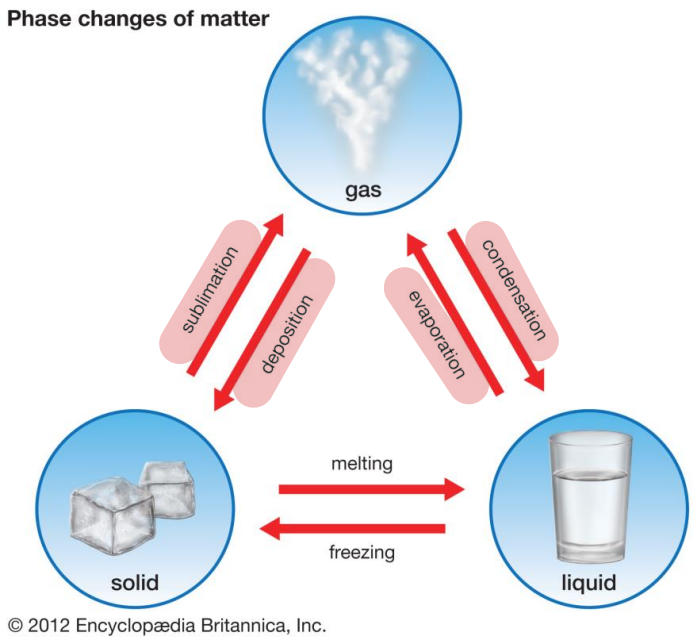


(18.32)

- Peak of each curve $v_{mp} = \sqrt{\frac{2kT}{m}}$ (18.34)
- Average speed $v_{av} = \sqrt{\frac{8kT}{\pi m}}$ (18.35)
- RMS speed $v_{rms} = \sqrt{\frac{3kT}{m}}$ (18.36)

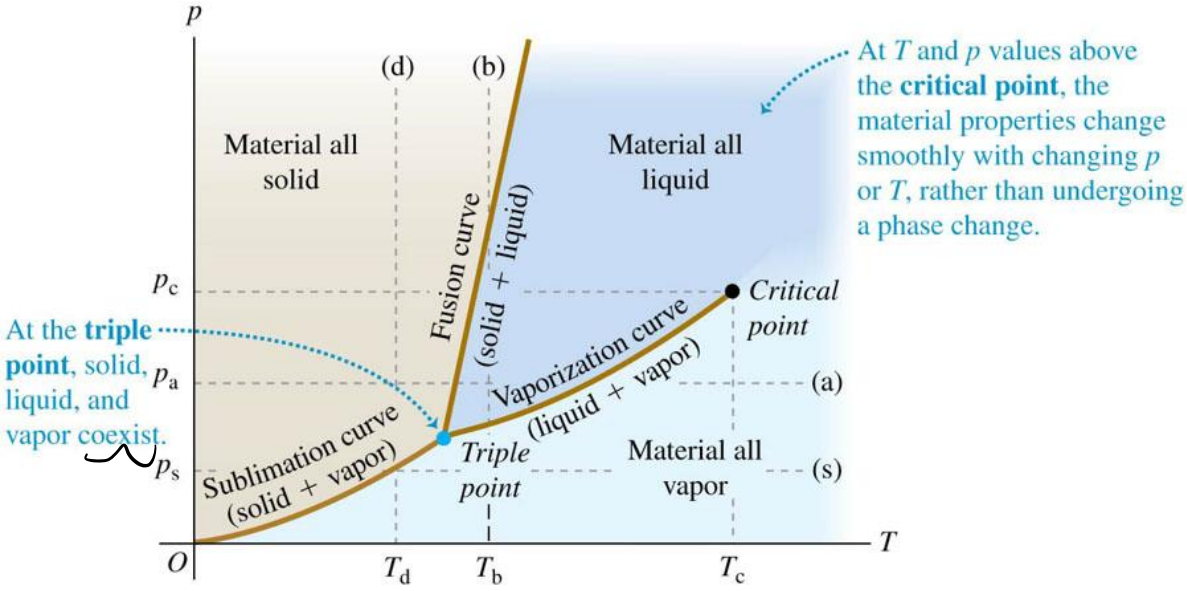
Phases of Matter

- For an ideal gas we ignore the interactions between molecules.
- But those interactions are what makes matter condense into the liquid and solid phases under some conditions.
  - Phase: the state of matter. Each phase is stable only in certain ranges of temperature and pressure.
  - Phase changes/Phase transitions/Phase Transformation: the change from one state to another state.



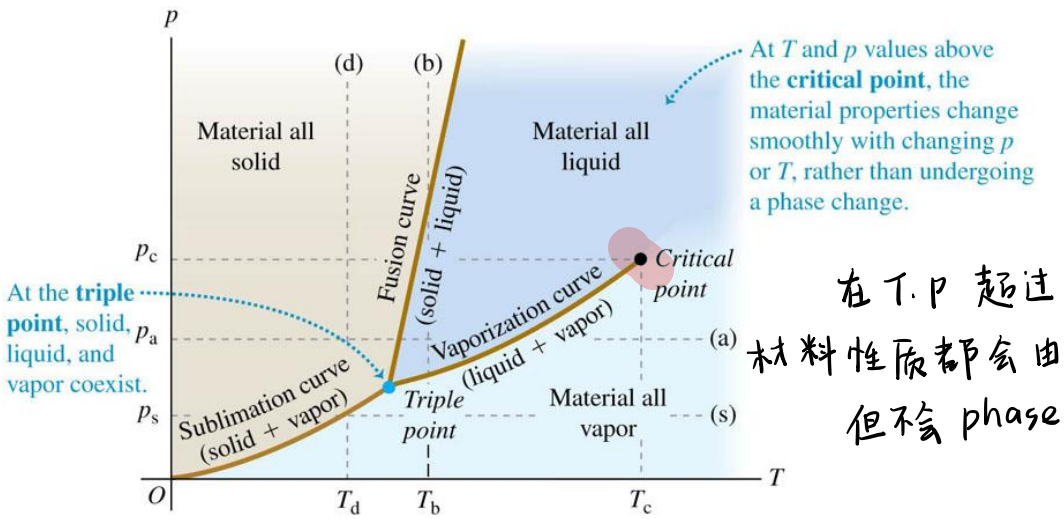
Phases of Matter

- A transition from one phase to another ordinarily requires phase equilibrium between the two phases, and for a given pressure this occurs at only one specific temperature. We can represent these conditions on a graph with axes p and T, called a phase diagram.
- Each point on the diagram represents a pair of values of p and T.



Phases of Matter

- Triple point: the point in which three phases coexist.
  - At any pressure less than the triple-point pressure, no liquid phase is possible.
- Critical point: the endpoint at the top of the vaporization curve.
  - — At T and p values above the critical point, the material properties change smoothly with changing p or T, rather than undergoing a phase change.



在T.P 超过 critical point.
材料性质都会由P.T的改变而改变
但不会 phase change

Phases of Matter

- Atmospheric pressure on earth is higher than the triple-point pressure of water.
- Depending on the temperature, water can exist as a vapor (in the atmosphere), as a liquid (in the ocean), or as a solid (like the iceberg shown here).



Summary

- Thermal Properties of Matter has been studied.
  - Basic concepts and principles are discussed with detailed analysis and examples
    - **1. Equations of State**
      - **1) State Variables 2) Ideal-Gas Equation**
      - **3) van der Waals Equation 4) pV-diagram**
    - **2. Molecular Properties of Matter**
      - **1) Molecules and Intermolecular Forces**
      - **2) Moles and Avogadro's Number**
    - **3. Kinetic-Molecular Model of An Ideal Gas**
      - **1) Collison and Gas Pressure**
      - **2) Pressure and Molecular Kinetic Energies**
      - **3) Collisions Between Molecules**
      - **4) Heat Capacities**
      - **5) Molecular Speeds / Maxwell-Boltzmann Distribution**
      - **6) Phases of Matter**
- Textbook
  - Chapter 18: Thermal Properties of Matter